

Numerical analysis of low-current arc characteristics in micro-TIG welding

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ABSTRACT

A semi-empirical two-temperature (2-T) arc model is developed to study the near-electrode regions and thermal non-equilibrium characteristics of low-current arcs that employ pin-to-pin electrodes in micro-tungsten inert gas (TIG) welding. Special emphasis is given to accurately depict the electrodes-arc attachment area and the spatial distributions of plasma temperature in 1–4 A argon arcs. The conditions of thermal non-equilibrium, as well as the coupling of the arc plasma and the cathode, are considered. The simulated electron temperature distributions are in good agreement with the arc appearances captured by the high-speed camera, and the predicted arc voltage is only 1.8 V lower than the experimental data. In contrast to TIG welding arcs that operate at higher currents, the temperature contours of the low-current arc column are parallel but slightly expanded in the vicinity of the electrodes. As the arc current decreases, the near-cathode voltage drop increases substantially. The electron temperature in the center of the arc column is almost constant, while the heavy particle temperature drops significantly, causing the arc column to contract. The heat flux at the center of the anode surface does not change as the arc current increases, but it increases significantly outside the 0.25 mm radius.

1. Introduction

Micro-tungsten inert gas (TIG) welding is derived from traditional TIG welding but uses a lower current of 0.5–20 A to melt the thin metal wire to create a micro-welded joint [1]. It is widely used to weld thermocouple wires, inductor coils, sensor wire ends, and varieties of other metal wires [2]. Unlike traditional TIG welding, the arc is generated in pin-to-pin electrodes rather than pin-to-plane electrodes with a lower current. The arc behavior between the conical electrode tip and the metal wire end is of vital importance since it strongly affects the formation of welded joints at the wire end. When dealing with arcs that are below 5 A, there is a higher risk of undesirable effects such as the movement of cathode spots and arc interruptions, which can lead to unstable arcs [3]. This instability can have a significant impact on the formation of micro-welded joints. Numerical simulation techniques have been utilized to study the properties of gas tungsten arcs, mainly due to the challenges associated with the experimental investigation [4,5]. Previously, the TIG arc plasma is assumed to be in the state of local thermodynamic equilibrium (LTE). However, this assumption is much less favorable in relatively low-temperature regions due to the difference in electron temperature (T_e) and heavy particles temperature (T_h) [6]. Therefore, many scholars developed a two-temperature arc model to investigate the thermal non-equilibrium effect [7], including the effects

of the external magnetic field [8] and shielding gas composition [9]. Later, it was discovered that the non-equilibrium characteristics become more pronounced and can be observed in the arc column as the arc current drops to a few amperes. Savovic et al. [10,11] measured the T_e and T_h of the steady argon arc and found that the ratio of T_e to T_h increases with the decreasing current. According to experimental results of Nick et al. [12], thermal non-equilibrium is observed in a 4 mm arc for electron number density (n_e) $< 6 \times 10^{22} \text{ m}^{-3}$. Benilov [13] found that the T_e of the arc column is as high as 7500 K for a 2.8 A arc, while the T_h is only 3000 K. Although thermal non-equilibrium has been considered in low-current arc modeling, the physical mechanism of arc column shrinkage in a low-current arc with decreasing current remains unclear.

Merely taking into account the thermal non-equilibrium characteristics is not sufficient for a thorough analysis of micro-TIG welding. The arc attachment on the cathode surface also has a significant effect on the characteristics of the low-current arc. In standard TIG welding arc models, the arc characteristics in the near cathode layer showed significant differences for cathode surfaces with different geometrical configurations [14–16]. For high-intensity discharge (HID) lamps, the diameter of the cathode is relatively small ($< 1.5 \text{ mm}$), and the near-cathode voltage drop U_c is sensitive to the attachment area. At a pressure of 0.2 MPa, the U_c of a 2 A arc is around 28 V with a cathode radius of 0.75 mm [17]. Schmitz and Riemann [18] developed the cathode

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boundary layer model of HID lamp, revealing that the U_c of the arc is mainly affected by the cathode radius rather than the gas pressure. Dabringhausen et al. [19] found that the U_c of the 2 A arc increases from 12 V to 17.5 V when the cathode radius is increased from 0.3 mm to 0.5 mm at a 0.26 MPa pressure in HID lamps. The near-cathode plasma layer model developed by Benilov and Cunha [20] shows that the cathode radius of the 1 A arc has a much greater influence on the U_c than that of the 2 A arc. This means that the cathode size has a significant impact on U_c . In comparison to the micro-TIG welding, the HID lamps in aforementioned experiments and models are operated at higher pressure with a cylindrical cathode to reduce heat flux density and enhance electrode life. In contrast, micro-TIG welding requires a higher heat flux density on the cathode surface to stabilize the arc [3]. As a result, their cathode geometry was quite different. Furthermore, it will inevitably result in quite different near-cathode layer characteristics, such as U_c , cathode surface temperature, heat flux density, and current density. Therefore, measuring the near-cathode voltage drop and developing a matching near-cathode layer model is essential for studying micro-TIG welding.

In addition, it has been discovered that the near-anode region of low-current arcs differs significantly from that of a conventional TIG welding arc. The latter model generally ignores the near-anode voltage drop U_a , as its magnitude is only between 1 and 2.5 V for a 25–200 A arc irrespective of the polarity of the voltage drop [21]. However, the U_a in low-current arcs exceeds this range significantly. Redwitz et al. [24] measured U_a in a HID lamp with an anode diameter of 1 mm using the Langmuir probe. They found that the U_a is equal to 8 V, 7 V, and 5 V for 1 A, 2 A, and 4 A arcs, respectively. Furthermore, the U_a increases with anode diameter. To better understand the energy transfer mechanism at the anode boundary, Mentel and Heberlein [25] established an anodic plasma boundary layer model consisting of a constriction zone, an anode sheath, and the anode solid without considering the arc column. However, the parameters of the model are derived from measured and evaluated values, not from the coupling of the arc column. The plasma properties near the anode are not well explained. Heberlein et al. [26] pointed out that the anode-arc attachment mode of high-intensity arcs has a great influence on the current and heat flux density at the anode surface. The above-cited experiments and models reflect that the U_a of low-current arcs cannot be ignored. In addition, HID lamps models focus on the energy balance and total heat flux input to the anode. These models do not provide the current density distribution on the anode surface. Instead, the average current density is used to calculate the power equation. Such a model is not suitable for micro-TIG welding because the current density distribution on the anode surface is extremely important to the welding quality of the micro-welded joint. Furthermore, because of the non-melting anode and higher operating pressures in HID lamps, it is questionable whether the constriction zone in the anode boundary layer persists in micro-TIG welding. Therefore, an optimized model that considers both U_a and current density distribution is required to describe the near-anode region characteristics in micro-TIG welding.

In our work, a high-speed camera is used to photograph 1–4 A arcs, especially the arc light near the anode. Then, the voltage drops in the near cathode and anode layers are derived by measuring the plasma potential along the arc axis. Based on the experimental results, a semi-empirical 2-T model of low-current arc in micro-TIG welding was established. The arc plasma is coupled to the cathode boundary layer to predict the U_c as well as the arc attachment at the cathode boundary. However, the U_a is not predicted by the arc model. Instead, the U_a obtained by experiments is adopted in the arc model to predict the near-anode arc characteristics. The temperature change of the arc column is presented as the current decreases from 4 A to 1 A to reveal the physical mechanism responsible for arc column shrinkage. The objective of this study is to provide detailed information about the arc characteristics in micro-TIG welding and to offer guidance for improving the formation of micro-welded joints.

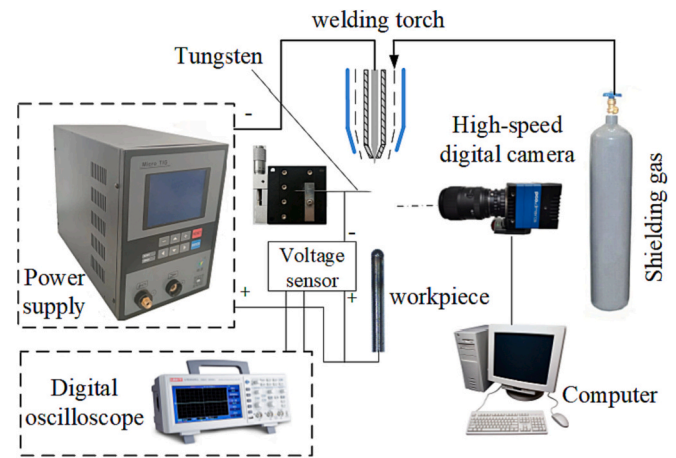


Fig. 1. Schematic diagram of the information acquisition system in micro-TIG welding.

2. Experimental procedures

2.1. Material and equipment

The tungsten cathode was WC20 (i.e., W + 1.8–2.2 wt%CeO₂) with a diameter of 1.0 mm and a taper angle of 30°. Pure argon was used as the shielding gas at the flow rate of 2 L/min, and the workpiece was a SUS304 steel wire with a diameter of 0.8 mm and its top was melted by an arc into a hemispherical shape. Long gap low-current arcs were obtained by a self-made micro-TIG machine. Its no-load voltage was 108 V and the rated output current was 20 A with a minimum output current of 0.1 A. A high-speed digital camera (PCO.Dimax CS2-1080p) was used to capture the arc appearances. The arc voltages were sampled with a hall voltage sensor (VSM025A) and a digital oscilloscope (DPO2024B, Tektronix, America).

2.2. Experimental methods

The schematic diagram of the data acquisition system in micro-TIG welding is shown in Fig. 1. For a clear observation of the difference between the arc column and the near-electrode regions, the electrode gap was set at 8 mm. When the current drops, a more pronounced contraction of the arc column can be observed. The large gap also helps to prevent the workpiece from being affected by high-temperature plasma near the cathode region. A high voltage arc initiation technique was used to ignite the arc. Moreover, an insulating ceramic tube was placed on the tungsten electrode to eliminate cathode spot motion. The experimental method is described in our previous arc ignition experiments [3]. The camera's aperture and exposure time were properly adjusted to clearly observe the arc profile and the arc light near the electrodes. A narrow-band optical filter (wavelength: 415 nm, half-width: 15 nm) was utilized to accurately observe the arc light while effectively avoiding the influence of other light sources. In the spectral distribution of TIG arc light, there are five spectral bands with stronger radiation intensity in the wavelength range of 400–430 nm [27].

To obtain the U_c and U_a , a tungsten probe with a diameter of 0.3 mm was inserted into the arc column to measure the plasma potential along the arc axis at defined distances d from the anode. The tip of the tungsten probe was polished to a cone angle of 20°. The other end of the tungsten probe was attached to a device that can be raised and lowered vertically with a 0.05 mm moving accuracy. The linear extrapolation of the plasma potential to a distance $d = 0$ gives the near-anode deviations from the linear potential drop [17]. These deviations are commonly defined to be the electrically measured near-anode voltage drop U_a . Similarly, the electrically measured near-cathode voltage drop U_c was obtained using

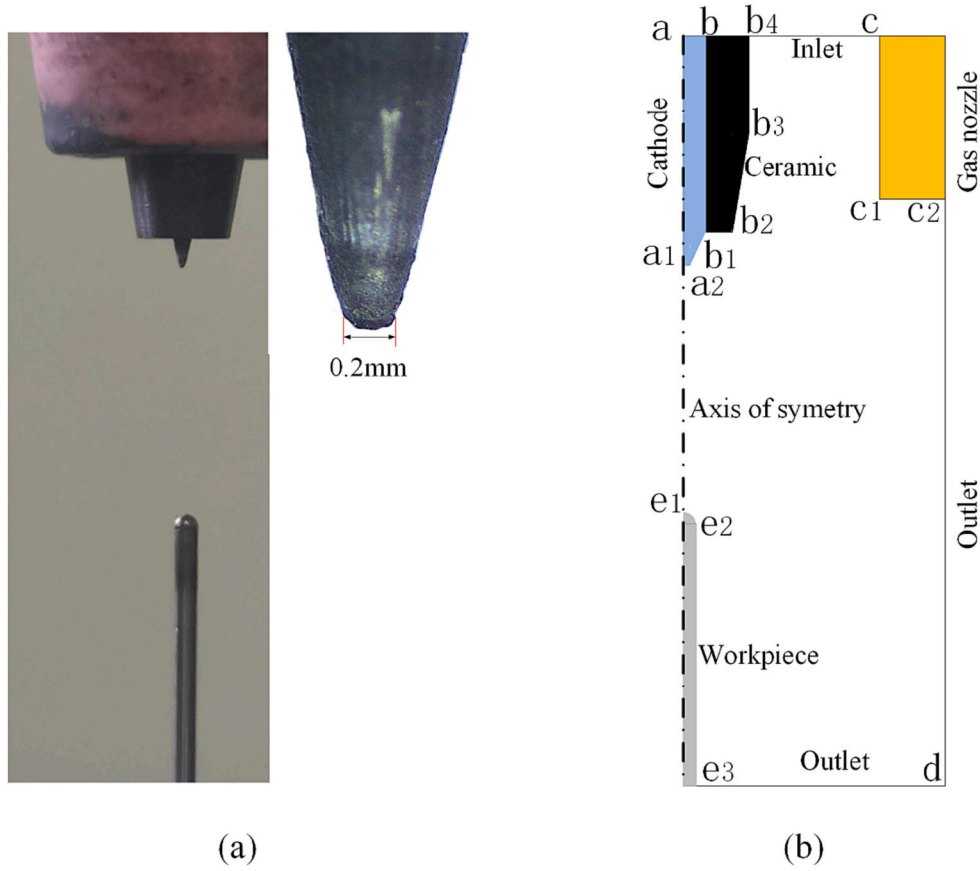


Fig. 2. Structure diagram of the welding torch and the workpiece: (a) actual welding structure and tungsten tip shape, and (b) computational domain of the model.

the same linear extrapolation method. To evaluate the effect of a tungsten tip insertion in the arc column on arc voltage, arc voltages measured with and without tungsten probe insertion were compared.

3. Two-temperature arc plasma model

3.1. Computational domains and main assumptions

The structure of the welding torch and workpiece in the actual welding process and an enlarged view of the tungsten tip are presented in Fig. 2(a). In our experiments, the cathode geometry influences the near-cathode voltage drop. Thus, a long-used tungsten electrode was adopted, whose geometry remains almost constant in low-current welding. In the model, the tungsten cathode is 1 mm in diameter and 7 mm in length, which has a fattened tip with a radius of 0.1 mm and a taper angle of 30°, as shown in Fig. 2(b). A ceramic tube is added to the cathode to simulate the actual situation. Only in this way, the cathode spot motion can be eliminated, and the long gap low-current arcs can be obtained in the welding experiments [3]. To simplify the calculation, the computational domains are arranged as axisymmetric. The computational domains include ceramic tube, solid cathode, and arc plasma regions. The arc is generated vertically, with the cathode on top and the workpiece below.

Since the model is designed to simulate arc characteristics, the melting of the workpiece is not considered. Thus, only the temperature of the anode surface is set without considering the heat transfer that occurs inside it. The model simulates a quasi-steady arc with the workpiece in a molten state. In this scenario, the arc voltage and shape exhibit little change, which allows for the arc to be considered approximately in a steady state. In accordance with the experiments, the anode upper surface is set to a hemispherical shape located 8 mm away from the cathode tip. In addition, the current density is set at the cathode

tip (“a1-a2-b1”) and the anode surface (“e1-e2-e3”). The purpose of these settings is to predict the cathode and anode arc attachment area. The specific details and physical mechanisms are described in Section 3.3. In order to investigate the characteristics of micro-arcs, the arc current is varied between 1 and 4 A.

The 2-T model is based on the following assumptions: (1) The micro-arc is steady at atmospheric pressure and the anode is non-ablative. (2) Argon is considered compressible and laminar. (3) A software named Fluent is used to solve the thermal non-equilibrium of arc plasma in the 2-T model. The quasi-neutral and chemical equilibrium conditions are satisfied, and only singly charged argon ions are considered. (4) Self-induced magnetic fields and gravity are considered. (5) The arc plasma is assumed to be optically thin to all radiation.

3.2. Governing equations

The 2-T model employs a set of partial differential equations to express required conservation equations on the basis of magnetohydrodynamic approaches. The stationary solutions of the 2-T model use the following equations in the arc region:

The mass conservation equation:

$$\nabla \cdot (\rho \mathbf{u}_g) = 0 \quad (1)$$

Here, ρ and $\mathbf{u}_g$ represent the mass density of plasma constituents and the gas flow velocity, respectively.

The momentum conservation equation:

$$\rho(\mathbf{u}_g \cdot \nabla) \mathbf{u}_g = \nabla \cdot \left(-p\mathbf{I} + \mu(\nabla \mathbf{u}_g + (\nabla \mathbf{u}_g)^T) - \frac{2}{3}\mu(\nabla \cdot \mathbf{u}_g)\mathbf{I} \right) + \mathbf{F}_L + \rho \mathbf{g} \quad (2)$$

Here, p is the sum of the partial pressures of various particles, which

is expressed as:

$$p = (n_a + n_i)k_B T_h + n_e k_B T_e \quad (3)$$

where n_a , n_i and n_e are the number densities of atoms, ions and electrons respectively. n_e is equal to n_i due to the charge neutrality. T_h and T_e are the temperatures of heavy particles and electrons respectively. k_B , μ , $\mathbf{g}$ and $\mathbf{I}$ are the Boltzmann constant, viscosity coefficient, gravity acceleration, and identity matrix, respectively. The Lorentz force $\mathbf{F}_L = \mathbf{j}_q \times \mathbf{B}$, where $\mathbf{j}_q$ is the total electric current density and $\mathbf{B}$ is the magnetic flux density.

The electron energy equation and heavy-species energy equation are obtained from [16]:

$$\nabla \cdot \left(\frac{5}{2} n_e k_B T_e \mathbf{u}_g \right) = \nabla (\kappa_e \nabla T_e) - e \mathbf{j}_e \cdot \mathbf{E} - \frac{5}{2} k_B (\mathbf{j}_e \cdot \nabla T_e) - Q_{e-h}^{el} - Q_{rad} \quad (4)$$

$$\nabla \cdot \left(\frac{5}{2} (n_a + n_i) k_B T_h \mathbf{u}_g \right) = \nabla (\kappa_h \nabla T_h) + e \mathbf{j}_i \cdot \mathbf{E} + Q_{e-h}^{el} \quad (5)$$

Here, κ_e and κ_h represent the thermal conductivities of electrons and heavy particles, which are obtained from Eqs. (16) and (17) in [28], respectively. $\mathbf{j}_e$ and $\mathbf{j}_i$ denote the density flux of electrons and ions, respectively. In the 2-T model, $\mathbf{j}_i = -e \mathbf{j}_e$ represent the total current density since electrons move much faster than ions in the arc region and the term $e \mathbf{j}_i \cdot \mathbf{E}$ in Eq. (5) is removed. The terms Q_{e-h}^{el} and Q_{rad} represent the energy transfer caused by elastic collisions and volumetric radiation, respectively. According to [14,28], Q_{e-h}^{el} and Q_{rad} are expressed as:

$$Q_{e-h}^{el} = \frac{3m_e}{m_i} n_e (\bar{\nu}_{ei} + \bar{\nu}_{ea}) k_B (T_e - T_h) \quad (6)$$

$$Q_{rad} = 2.6 \times 10^{25} \frac{p}{1 \text{ atm}} \left(\frac{1}{T_e} \right)^{2.52} \exp \left(- \frac{1.69 \times 10^5}{T_e} \right) \quad (7)$$

Here, m_e and m_i are the mass of the electron and ion respectively, $\bar{\nu}_{ei}$ and $\bar{\nu}_{ea}$ are the collision frequencies of an electron with ions and atoms respectively which are obtained from [29], $\Delta \epsilon_i$ is the energy loss due to inelastic collisions. Assuming chemical equilibrium and ignoring the secondary ionization of argon, the particle number densities n_e , n_i and n_a are solved by Eq. (3) and Saha equation, which is expressed as [22]:

$$\frac{n_e n_i}{n_a} = \frac{2Z_i}{Z_a} \left(\frac{2\pi m_e k_B T_e}{h^2} \right)^{1.5} \exp \left(- \frac{\epsilon_i}{k_B T_e} \right) \quad (8)$$

where Z_i and Z_a are the partition function of the ions and the neutral atoms, respectively, h is Plank's constant and $\epsilon_i = 15.75 \text{ eV}$ is the ionization energy.

The current continuity equation and Ohm's law:

$$\nabla \cdot \mathbf{j}_i = 0 \quad (9)$$

$$\mathbf{j}_i = -e \mathbf{j}_e = \sigma \mathbf{E} = -\sigma \nabla \phi \quad (10)$$

The Maxwell's equation:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{j}_i \quad (11)$$

Here, $\mathbf{E}$, σ , μ_0 and ϕ represent the electric field strength, electrical conductivity, permeability of free space and electric potential, respectively, $\mathbf{A}$ is the magnetic vector and is obtained from $\mathbf{B} = \nabla \times \mathbf{A}$. The conductivity σ is obtained by interpolation method from [31], which was determined by T_e and T_h .

The energy and charge conservation equations in the cathode solid are given as:

$$\nabla \cdot (\kappa_c \nabla T) + \sigma_c E^2 = 0 \quad (12)$$

$$\nabla \cdot \sigma_c (-\nabla \phi) = 0 \quad (13)$$

The energy conservation equation in the ceramic solid is written as:

Table 1

Boundary conditions considered in the 2-T model.

Boundary	T, T_h	T_e	p	u_g	ϕ	A
a1-a2-b1	$T_h = T_c$	q_{ec}	–	0	$j = -j_r(\text{plasma})$ $j = j_r(\text{cathode})$	$\frac{\partial A}{\partial n} = 0$
b1-b2-b3-b4	$T_h = T_{ce}$	$\frac{\partial T_e}{\partial n} = 0$	–	0	$\frac{\partial \phi}{\partial n} = 0$	$\frac{\partial A}{\partial n} = 0$
a-b	1400 K	–	–	–	0	–
b-b1	coupled	–	–	–	$\frac{\partial \phi}{\partial n} = 0$	$\frac{\partial A}{\partial n} = 0$
b4-c	300 K	1000 K	1 atm	$\frac{Q}{S}$	$\frac{\partial \phi}{\partial n} = 0$	$\frac{\partial A}{\partial n} = 0$
c-c1-c2	$\frac{\partial T_h}{\partial n} = 0$	$\frac{\partial T_e}{\partial n} = 0$	–	0	$\frac{\partial \phi}{\partial n} = 0$	$\frac{\partial A}{\partial n} = 0$
c2-d-e3	$\frac{\partial T_h}{\partial n} = 0$	$\frac{\partial T_e}{\partial n} = 0$	$\frac{\partial p}{\partial n} = 0$	Open	$\frac{\partial \phi}{\partial n} = 0$	$\frac{\partial A}{\partial n} = 0$
e3-e2-e1	$T_h = T_a$	q_{ea}	–	0	j_{ea}	$\frac{\partial A}{\partial n} = 0$

$$\nabla \cdot (\kappa_{ce} \nabla T) = 0 \quad (14)$$

Here, κ_c and κ_{ce} represent the thermal conductivity of the cathode and ceramic solid respectively, σ_c is the electrical conductivity of the cathode solid. The ceramic solid is assumed to have no current conduction.

3.3. Boundary conditions and calculation method

Table 1 shows the boundary conditions based on the actual conditions shown in Fig. 2(b). Where S represents the area of the inlet boundary (“b4-c”). At the inlet (“b4-c”), the heavy particle temperature is set to only 300 K due to the low-current arcs. On the axis of symmetry (“a-a1-e1”), the radial gradients of all dependent variables are set to zero. The temperature on the boundary (“a-b-b4”), which is 7 mm away from the cathode tip, is set to 1400 K based on the temperature variation of the tungsten cathode under similar current conditions as described in [17]. The heavy particle temperature at the interface between the cathode solid, ceramic solid, and plasma is consistent with the surface temperature of the cathode (T_c) and ceramic solid (T_{ce}). The cathode material is tungsten containing 2 % cerium oxide [29]. The shielding gases were pure argon with a flow rate Q of 2 L/min.

The interaction between the cathode and the near-cathode plasma layer at the cathode boundary (“a1-a2-b1”) is described using a non-linear surface heating model, which is a typical approach for modeling the plasma boundary layer of high-intensity discharge (HID) cathodes [32]. According to [23], the total current density j_t and power flux density q_c across the boundary layer are described as:

$$j_i = j_{em} - j_{bd} + j_i \quad (15)$$

$$q_c(T_e, U_c) = j_i U_c - \frac{j_i}{e} W_{eff} - 2.5 \frac{j_i}{e} k_B T_e - q_{rad} \quad (16)$$

The total energy flux of electrons transferred from the near-cathode layer to the ionization layer-plasma interface is described as [33]:

$$q_{ec} = \frac{j_{em}}{e} (2k_B T_c + e U_{cs} - \Delta W) + W_e - \frac{j_{bd}}{e} (2k_B T_e + e U_{cs} - \Delta W) - E_i \frac{j_i}{e} \quad (17)$$

Here, $j_{em} = e A_r T_c^2 \exp \left(- \frac{W - \Delta W}{k_B T_c} \right)$ denotes the density of thermionic electron emission current for $T_c > 2000 \text{ K}$, the electron emission current is ignored when $T_c \leq 2000 \text{ K}$ [32]. $j_{bd} = \frac{e}{4} \sqrt{\frac{8k_B T_e}{\pi m_e}} n_{es} \exp \left(- \frac{e U_{cs}}{k_B T_c} \right)$ represents the electrons flow density diffused back to the cathode and $j_i =$

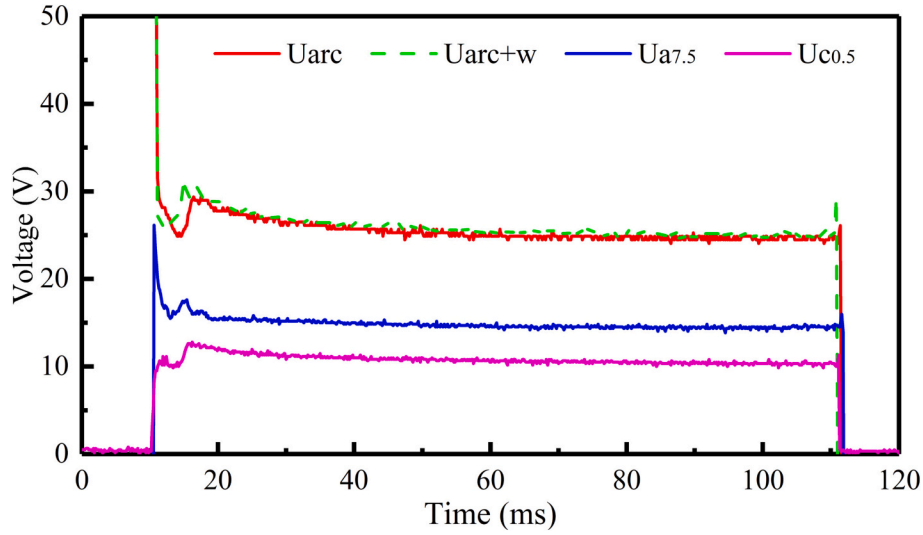


Fig. 3. Waveforms of arc voltage and the voltage drop on the arc axis 7.5 mm away from the anode for a 4 A arc.

$en_{es}\sqrt{k_B(T_c + T_e)/m_i}$ accounts for the ions current density transferred to the cathode. q_{rad} , A_r , W_{eff} and ΔW account for the radiation energy losses, Richardson's constant, the effective work function, and the Schottky reduction, respectively [34]. The near-cathode layer is denoted as the space-charge sheath and the ionization layer adjacent to the arc column. Their voltage drops are U_c , U_{cs} and U_{ci} , respectively, and $U_c = U_{cs} + U_{ci}$. In our model, the U_c along the cathode surface is considered constant due to the narrowed cathode-arc attachment area [35]. $U_{ci} = \frac{k_B T_e}{e} \ln \frac{n_{ei}}{n_{es}}$ is obtained from [36], where n_{ei} and n_{es} are the electron number densities on the side of the ionization layer adjacent to the arc column and the space charge sheath, respectively. The U_c is taken as an input parameter to obtain a heat flux to the cathode. It is then adjusted to increase or decrease the heat flux density to ensure that the integration of the current j_t is consistent with the target current [34]. The U_{cs} is evaluated as: $U_{cs} = U_c - U_{ci}$. E_c , E_i and W_e are the electric field strength of the space charge sheath, the ionization energy and the energy gained by electrons in the ionization layer, respectively. At the arc-cathode interface, the heat flux transported to the cathode metal is set to $q_c(T_e, U_c)$; the heavy-particle temperature is set to be equal to cathode surface temperature T_c .

At the anode boundary ("e1-e2-e3"), some simplifications are made. That is, only the electron current density from the plasma to the anode j_{ea} is considered. Because the anode surface temperature is relatively low and the measured near-anode voltage drops were higher than 5 V in experiments. Such a high voltage drop can decelerate ions from the arc column so strongly that the ion current reaching the anode is negligibly small [23]. The near-anode layer is also divided into the space-charge sheath and the ionization layer. Their voltage drops are U_a , U_{as} and U_{ai} , respectively, and $U_a = U_{as} + U_{ai}$. In our model, the U_a is constant along the anode boundary. In most two-temperature arc models, the anode surface potential is usually set to 0. But this setting is not suitable for low-current arcs in pin-to-pin electrodes. Because the T_e profile obtained by this setting differs significantly from the arc profile obtained by a high-speed camera. Considering that the current on the anode surface is mainly formed by the electrons from the arc column entering the anode, the current density j_{ea} on the surface, analogous to Eq. (6) in [25], is written as:

$$j_{ea} = \frac{e}{4} \sqrt{\frac{8k_B T_e}{\pi m_e}} n_{ei} \exp\left(-\frac{eA}{k_B T_e}\right) \quad (18)$$

where n_{ei} is the electron number density at the interface between the ionization layer and the plasma side, A is an adjustable parameter used

to ensure that the integral of the j_{ea} over the anode surface is equal to the target current. In the process of solving the model, an initial anode surface current density distribution, the integral of which over the anode surface is equal to the target current, is given as the input parameters to get a converged solution first. Then, the predicted current density distribution j_{ea} is obtained from the previous solution. By adjusting parameter A , the integral of the predicted j_{ea} over the anode surface is made equal to the target current. Then, this new j_{ea} is used as the boundary condition of the current density on the anode surface. The preceding steps are repeated until the predicted current density matches the set current density. The formula is based on the physical principle that current enters the anode from the areas with higher n_{ei} and T_e . This setting has the following limitations: (1) The anode surface is almost not affected by the cathode jet (low-current arcs). (2) The arc is in a steady state. (3) The anode surface temperature is high enough. (4) The direction of the gas flow is parallel to the arc column.

The Electron energy balance equation in the anodic ionization layer, analogous to Eq. (4) in [33], is described as:

$$\frac{j_{ea}}{e} (2k_B T_e - eU_{as}) = j_{ea} U_{ai} - k_e \vec{\nabla} T_e \cdot \vec{n} - 2.5k_B T_e \vec{j}_s \cdot \vec{n} \quad (19)$$

Here, the left side of the equation represents the energy flux carried away by electrons leaving the ionization layer for the space-charge sheath. The terms on the right side represent energy gained by electrons in the ionization layer and the energy transferred from the arc column to the ionization layer, respectively. Thus, according to Eq. (4), the total energy flux density carried by the electrons from the near-anode layer to the ionization layer-plasma interface is derived as:

$$q_{e,a} = - \left(-k_e \vec{\nabla} T_e \cdot \vec{n} - 2.5k_B T_e \vec{j}_s \cdot \vec{n} \right) = \frac{j_{ea}}{e} (eU_a - 2k_B T_e) \quad (20)$$

As U_a is difficult to predict using a two-temperature model, it can only be obtained through experiments. Actually, the voltage drop of the plasma simply represents the arc column voltage drop (U_{col}) in the 2-T model. The total arc voltage U_{arc} is composed of the U_c , U_{col} , and U_a , i. e. $U_{arc} = U_c + U_{col} + U_a$. Our model is consistent with the experimental situation when the arc reaches stability, while the upper surface of the workpiece is in a molten state. Therefore, the heavy particle temperature T_a at the upper surface of the anode ("e1-e2") is set as the melting point of the workpiece material, i.e. 1450 K. On its side ("e2-e3"), the heavy particle temperature T_a drops linearly from the melting point at "e2" to room temperature at "e3". In fact, there is an additional heat flux $q_a = j_{ea}(2k_B T_e + W_a)/e - q_{rad}$ from the arc column to the anode [37]. W_a is the work function of the anode material. Therefore, the total energy balance

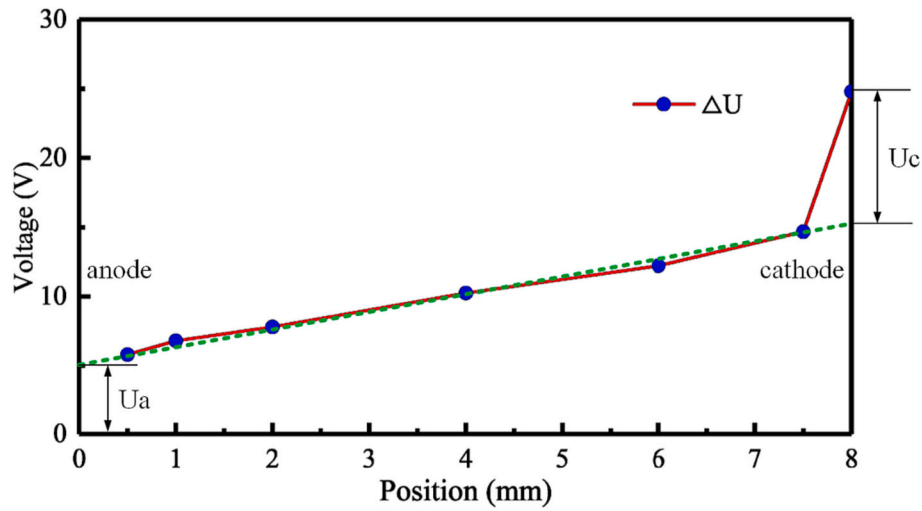


Fig. 4. Voltage drop ΔU between the anode and the plasma along the arc axis for a 4 A arc.

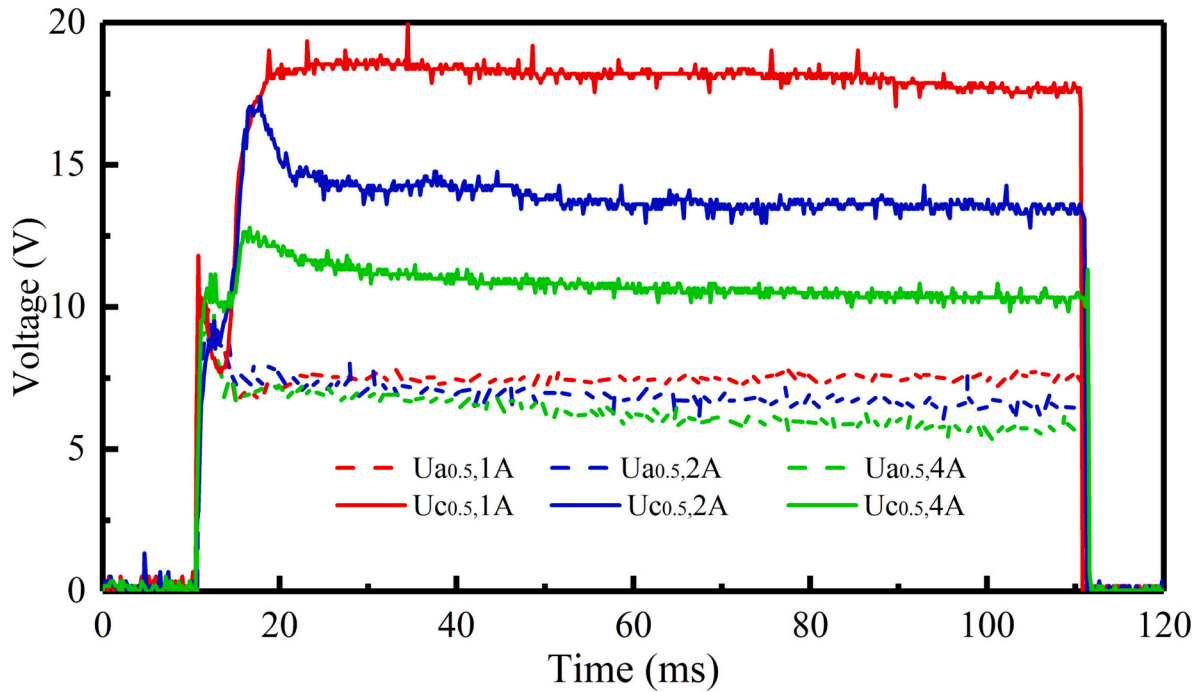


Fig. 5. Waveforms of the voltage drop on arc axis 0.5 mm from the electrodes for various arc currents.

equation, similar to that in [38], can be obtained:

$$q_{e,a} + q_a = \frac{j_{ea}}{e} (eU_a + W_a) - q_{rad} \quad (21)$$

The 2-T model is numerically solved by SIMPLEC algorithms on the FLUENT computing platform. For the steep parameter gradients in these regions, the meshes in the vicinity of the cathode and anode are divided into 200 and 80 sections, respectively. In the fluid domain, the initial T_e and T_h are set to 10,000 K and 2000 K, respectively. The transport properties of argon were calculated as described by Zhang et al. [30].

4. Results and discussions

4.1. Results of the experiments and the 2-T model

Fig. 3 shows the measured voltage waveform of a 4 A arc. $U_{a7.5}$

represents the voltage drop between the anode and the arc column plasma located 7.5 mm away from the anode, while $U_{c0.5}$ represents the voltage drop between the cathode and the arc column plasma 0.5 mm away from the cathode. U_{arc} and U_{arc+W} represent actual arc voltage and arc voltage with a tungsten probe inserted into the arc axis 0.5 mm away from the cathode, respectively. The results show that 60 ms after the ignition, the measured voltage remains stable, indicating that the arc has reached stability. In our previous work [3], high-speed photography of the arc also showed that the arc appearance remained almost the same during this period. The 2-T arc model is established based on the stable arc. In addition, U_{arc+W} is almost identical to U_{arc} , indicating that tungsten insertion in the arc column has little effect on arc voltage. The arc length is 8 mm, and U_{arc+W} is equal to the sum of $U_{a7.5}$ and $U_{c0.5}$. This means that inserting a tungsten probe into the arc column to measure the electric potential of the arc column plasma does not result in a large error.

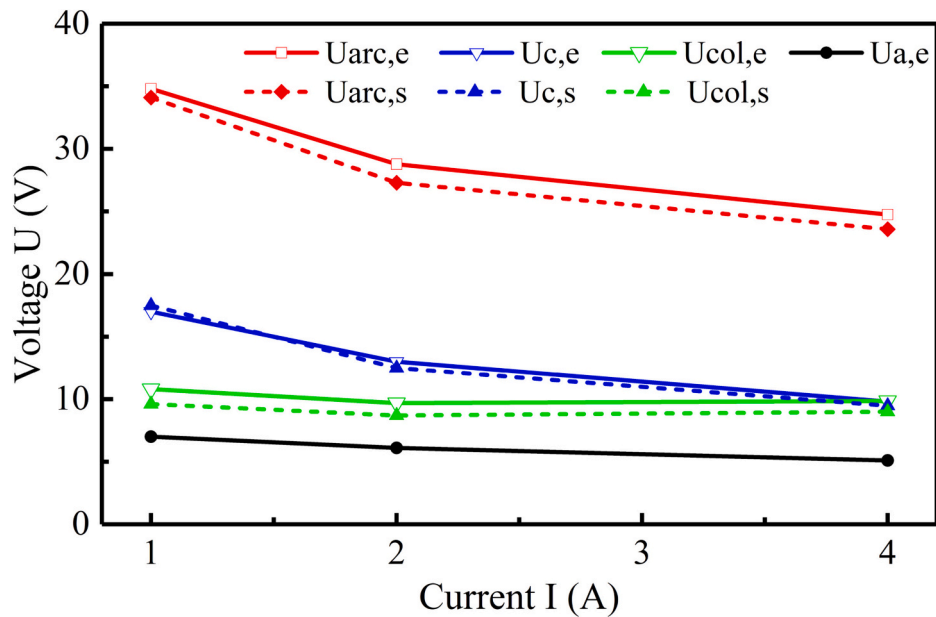


Fig. 6. Comparison of simulated and measured arc voltage for various arc current.

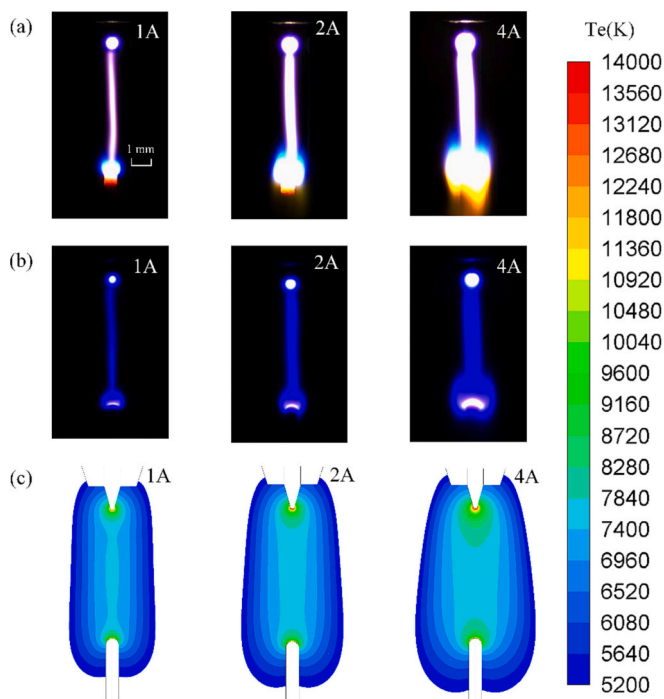


Fig. 7. Comparison of actual arc appearance and simulated T_e distribution with various currents: (a) exposure time 200 μ s without filtering light, (b) exposure time 500 μ s with an optical filter of 415 ± 15 nm wavelength, and (c) simulated T_e distribution.

As shown in Fig. 4, the voltage drop ΔU between the anode and arc column plasma is measured at different positions based on the feasibility of the above-given method. The U_c and U_a are determined by a linear extrapolation of ΔU toward the electrodes. The U_a is also used for anode boundary conditions of T_e in the 2-T model. Fig. 5 shows the waveforms of $U_{c,0.5}$ and $U_{a,0.5}$ for arcs with various currents. The implication of $U_{a,0.5}$ is consistent with $U_{a,7.5}$. For arcs ranging from 1 to 4 A, the key potential variation in the near electrode layer (0.5 mm) tends to stabilize after 60 ms of arc ignition. Therefore, it is reasonable to use the linear

extrapolation method to determine the U_c and U_a values for arcs ranging from 1 to 4 A.

Fig. 6 shows a comparison of predicted current-voltage characteristics with the measured value. The subscripts s and e stand for simulation and experiment, respectively. As presented in Fig. 6, the $U_{arc,s}$ agrees well with the $U_{arc,e}$, and both significantly decrease as the current increases. The primary factor contributing to the drop in arc voltage with decreasing arc current is the near-cathode voltage drop, which represents the majority of the total arc voltage. When the current increases from 1 A to 4 A, the $U_{c,s}$ drops from 17.5 V to 9.6 V, while the voltage of the arc column $U_{col,s}$ drops only 0.8 V. In addition, the near-anode layer has a non-negligible voltage drop $U_{a,e}$, which is >5 V for arcs below 4 A. For the 2-T arc model, it is challenging to include the near anode layer in the arc plasma simulation. At present, there is no suitable method to predict the U_a in the 2-T model. The measured U_a is added to the model for more realistic near-anode characteristics. Therefore, our 2-T arc model is semi-empirical rather than self-consistent, but it can also describe near-anode arc characteristics that are difficult to obtain experimentally. In general, the model provides a better understanding of the electrical and thermal characteristics of the micro-arc in the near-cathode layer, arc column plasma and near-anode layer.

Fig. 7 presents the photographs of the 1, 2, and 4 A arcs and the distributions of T_e in the 2-T model. High T_e means more collisions of electrons with heavy particles, resulting in more photons [39], so the distribution of T_e is comparable to the arc profile. As can be seen, the plasma shapes of the 2-T model match well with actual arc appearances. As the current decreases, the diameter of the arc column shrinks, in line with the shrinkage of the T_e profile. In addition, the 7400 K isotherm of T_e and the actual arc appearance on both sides of the arc column are parallel to the arc axis and are slightly expanded near the electrodes. When the optical filter (415 ± 15 nm) is used, the brighter white arc light near the electrode is clearly observed, while the arc column plasma appears blue light. This is due to the large amount of energy deposited into the near cathode and anode layers, resulting in a much higher T_e in these two regions than in the arc column plasma. In particular, the white arc light in the near-anode region of the 1–2 A arcs is quite thin. Unlike the HID lamp, there is no clear constriction zone in the near anode area of micro-TIG welding [24]. Consequently, it is logical to consider the near-anode layer as one-dimensional in the 2-T arc model.

The setting of the anode boundary condition in micro-TIG welding

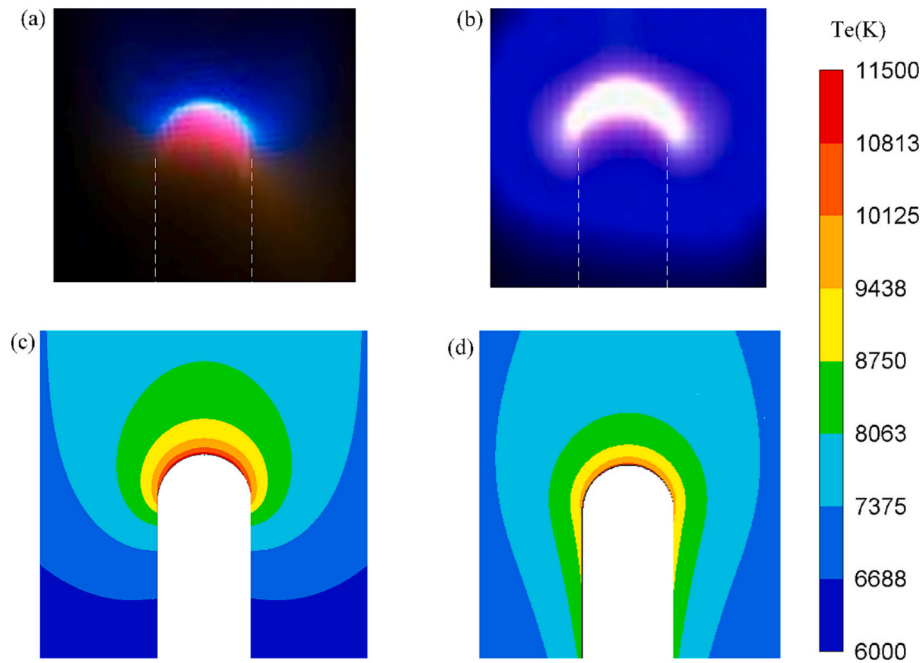


Fig. 8. Actual arc appearance and simulated T_e distribution near the anode for a 4 A arc: (a) exposure time 3 μ s without filtering light, (b) exposure time 500 μ s with an optical filter of 415 ± 15 nm wavelength, (c) set the anode surface current density to j_{ea} , and (d) set the anode surface electric potential to 0.

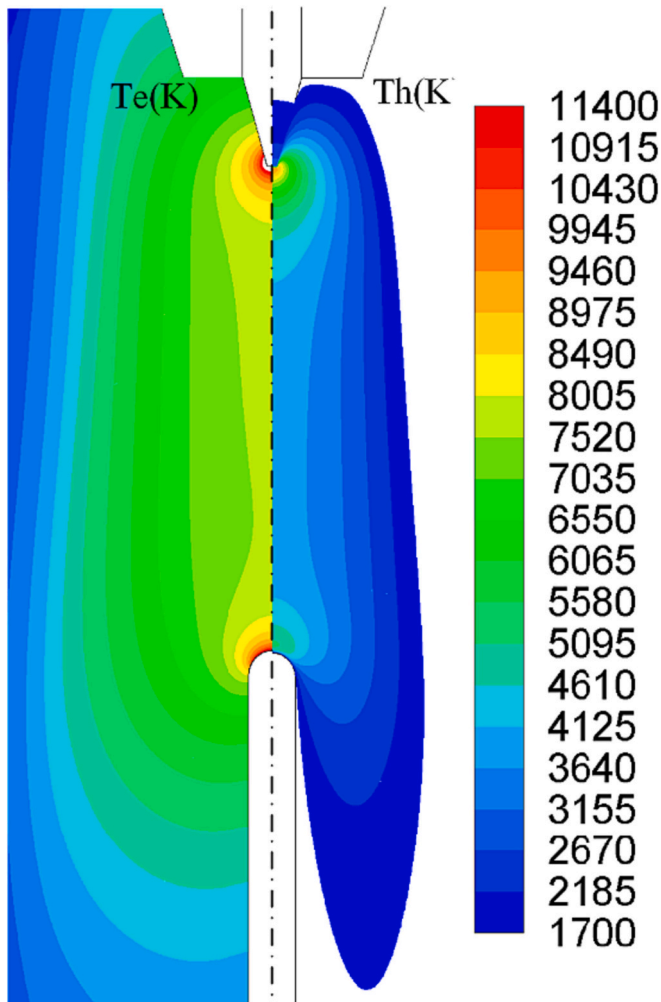


Fig. 9. Profile of the T_e and T_h for a 2 A arc.

differs from that of the traditional TIG welding in the 2-T arc model. As shown in Fig. 8(a–b), the brighter arc light appears only on the hemispherical surface of the anode. Away from the hemispherical surface of the anode, the arc light is relatively dim, indicating a lower T_e . The blue light is observed over the anode surface. It is caused by a combination of metal vapors (including iron, chromium, nickel, manganese, etc.) and argon. These vapors may have some effect on the near-anode arc characteristics. In our model, the near-anode voltage drop was measured experimentally and then applied to the model to account for the energy deposited by electrons in the near-anode layer. The measured near-anode voltage drop roughly includes the combined effect of the metal vapors. As shown in Fig. 8(d), if the electric potential on the anode surface is set to 0, the hot zone of T_e extends from the hemispherical surface to the cylindrical side, which is completely inconsistent with the arc light profile. This may be due to the existence of the space-charge sheath and the ionization layer. When current density j_{ea} is set on the anode surface, the profile of T_e is in good agreement with the arc brightness. Therefore, when establishing the boundary conditions of Ohm's law (Eq. (9)) for cylindrical anodes, it is more preferable to consider the entry of electrons from the arc column into the anode (Eq. (18)) as the relevant physical phenomenon.

4.2. Thermal non-equilibrium characteristics of low-current arc

Fig. 9 shows the distribution of T_e and T_h for a 2 A arc. In contrast to the high-current arcs, the thermal non-equilibrium phenomenon, i.e., the large deviation between the T_h and T_e , is observed throughout the arc region, including the near-cathode region. Although the maximum values of T_e and T_h are located close to the cathode tip, there is still a significant deviation between the two maximum values. Both T_e and T_h exhibit elongated cylindrical shapes within the arc column region. This is almost irrelevant to the electromagnetic force because the electromagnetic force is extremely small. The arc shape is almost unaffected when the F_L term in Eq. (2) is removed. According to Eqs. (4) and (5), the energy gained by the electrons in the arc column is primarily due to Joule heating, while the energy of the heavy particles is due to the collision with the electrons. For low-current arcs, the current in the arc column is concentrated in a narrow area, resulting in a columnar Joule

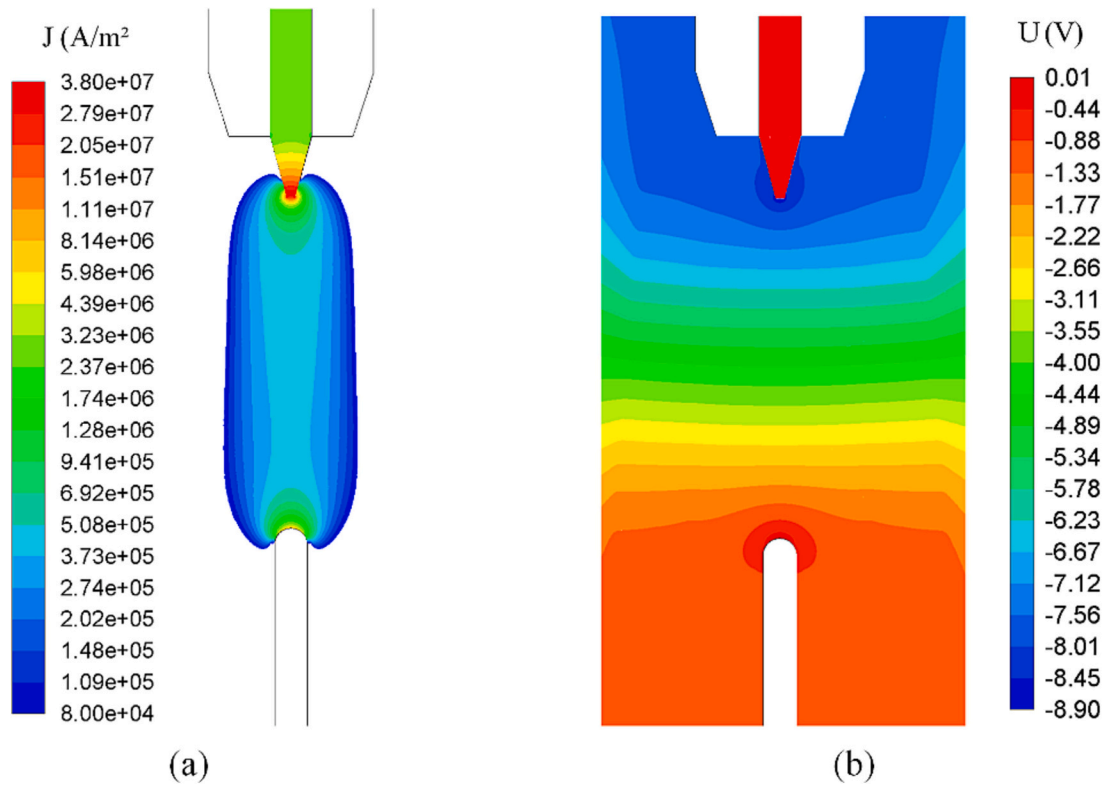


Fig. 10. Distribution of the current density (a) and electric potential (b) for a 2 A arc.

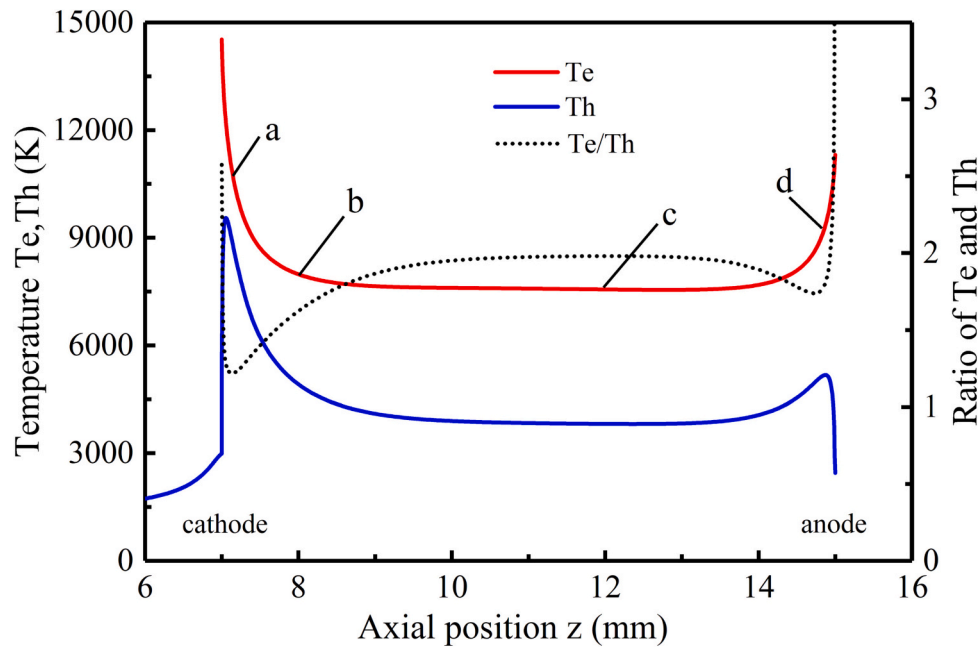


Fig. 11. Distributions of T_e , T_h and T_e/T_h along the symmetry axis for a 2 A arc.

heating distribution, i.e. an elongated temperature profile in the arc column.

Fig. 10 shows the distribution of the current density J_t and electric potential U for a 2 A arc. Since the electric potential is set to zero at the top of the cathode, the electric potential in Fig. 10 only represents the potential deviation in the respective regions rather than the actual electric potential. The current density of the arc column is mainly concentrated in the region of a 1.5 mm radius. The current density

within the radius of 0.4 mm is above $3.73 \times 10^5 \text{ A/m}^2$, while outside the region, the current density drops rapidly to $8.0 \times 10^4 \text{ A/m}^2$. The rapid decrease in current density is attributed to the rapid decrease of electric conductivity at the arc fringes. The contour lines of the electric potential in the arc column are mostly perpendicular to the arc axis, suggesting that the radial current density is negligible. The axial electric field intensity remains consistent at approximately 1.1 V/mm. In the region adjacent to the anode, the potential converges toward the anode,

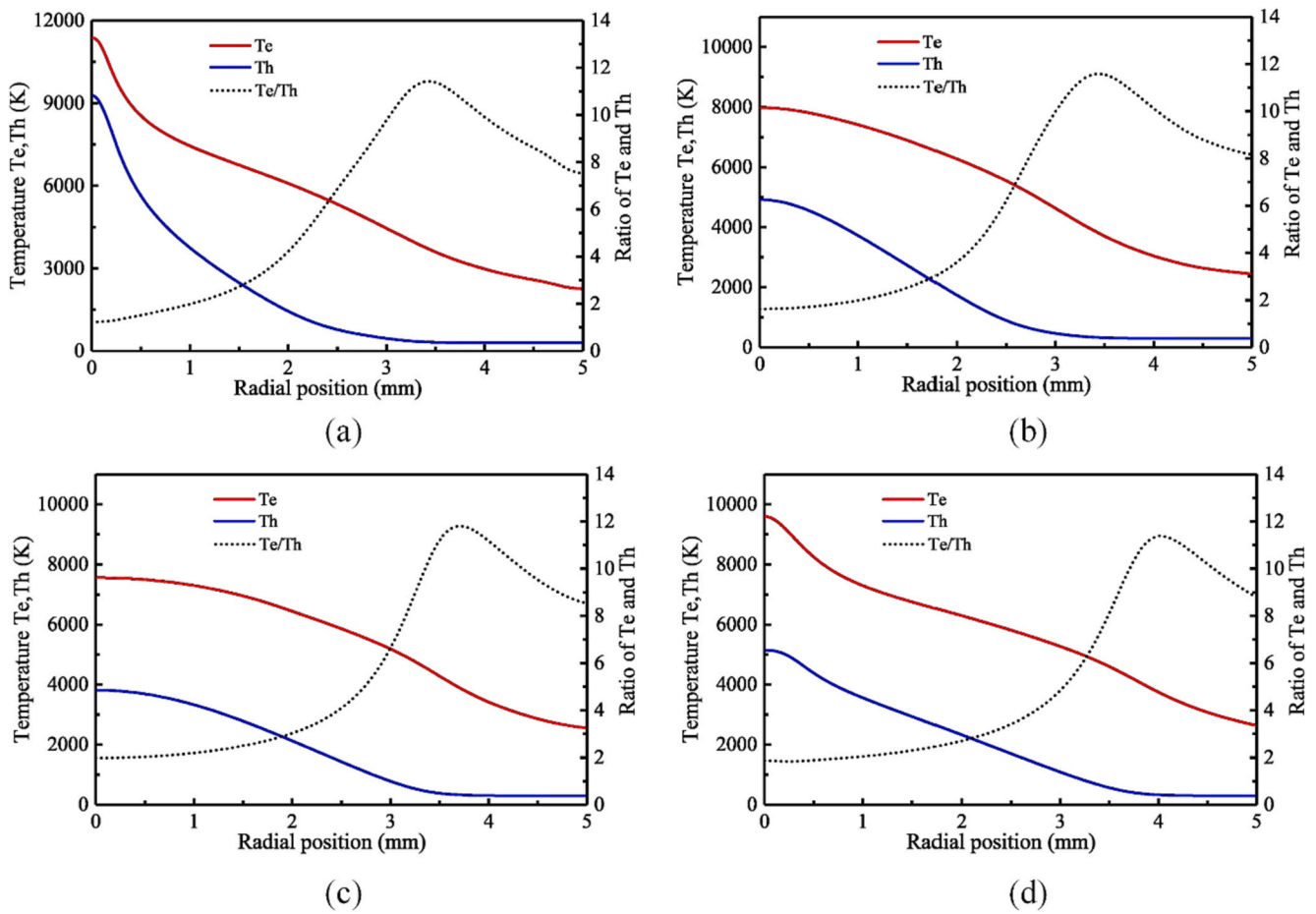


Fig. 12. Radial distributions of T_e and T_h for the axial position of 0.1 mm (a), 1 mm (b), 5 mm (c) and 7.9 mm (d) away from the cathode tip.

causing the current of the arc column to eventually converge on the anode surface. As presented in Fig. 6(a), almost all of the current enters the anode from its hemispherical surface, while the current entering the anode from its cylindrical side is negligible.

Fig. 11 shows the axial distributions of T_e and T_h along the arc axis, with the ratio $\theta = T_e/T_h$ describing the degree of deviation from thermal equilibrium. Along the axis, the T_e drops sharply within 1 mm of the cathode tip, followed by a smooth transition in the arc column, and

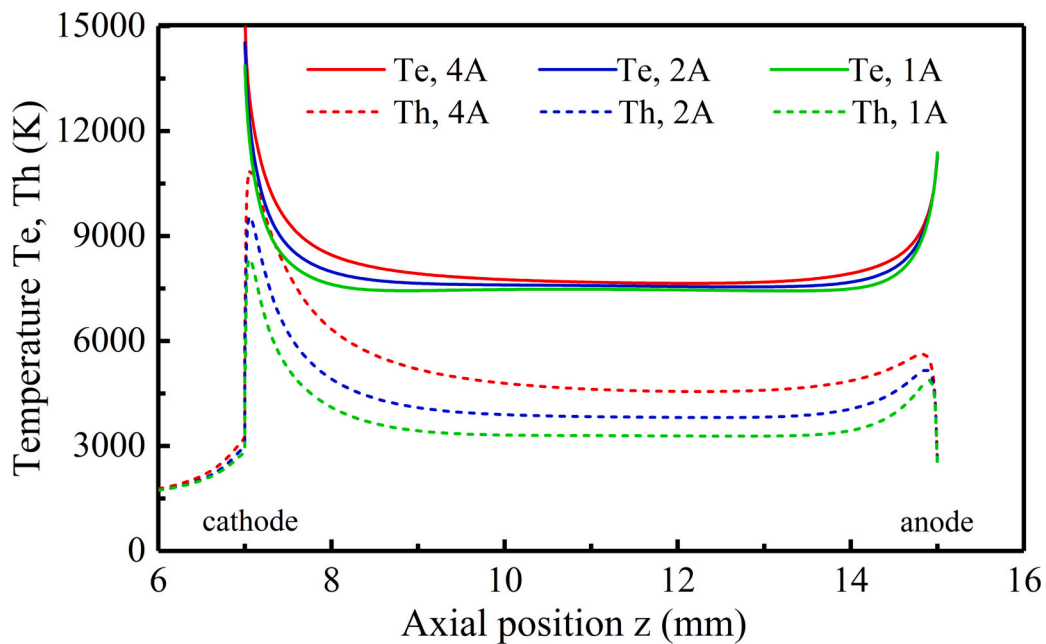


Fig. 13. Distributions of T_e and T_h along the symmetry axis for various arc currents.

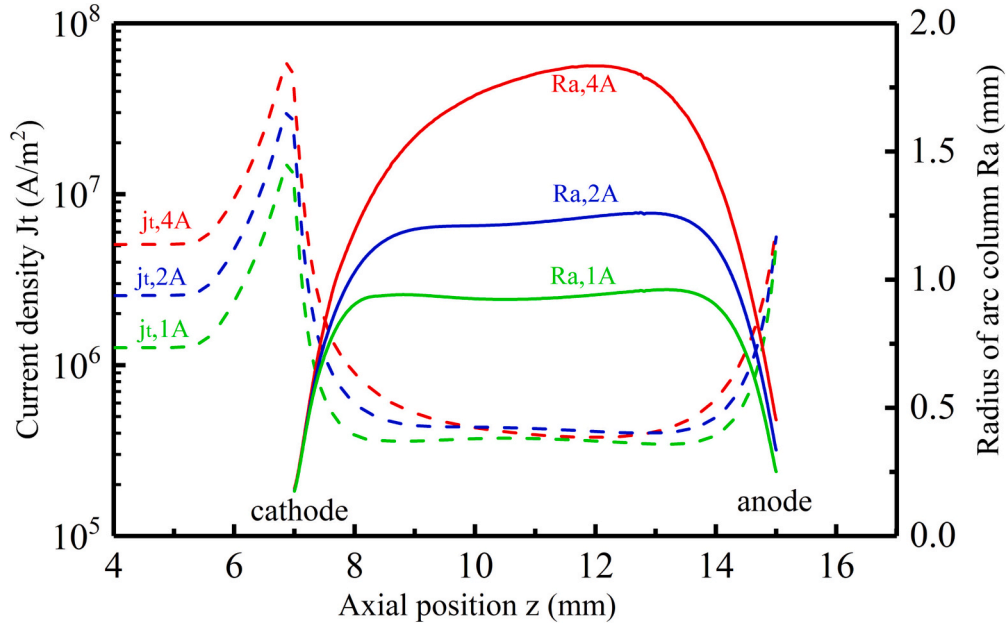


Fig. 14. Distributions of current density and arc radius along the symmetry axis for various currents.

finally a rapid increase adjacent to the anode. In near cathode and anode regions, the maximum T_e reaches 14,500 K and 11,400 K, respectively. In addition, there are two T_h peaks along the arc axis, which appears at 0.07 mm below the cathode tip (9530 K) and 0.14 mm above the anode (5170 K). Similarly, there are two minima of θ on the arc axis, 1.22 ($z = 7.138$ mm) and 1.74 ($z = 14.725$ mm). The T_h in the arc column is only 3810 K due to the low energy transferred to the heavy particles because the electron temperature is only 7550 K. Therefore, the coefficient θ of non-LTE reaches 1.98 even in the arc column. Although the T_h of the arc column is low, the T_e is large enough to generate sufficient electrons number density to maintain high conductivity, thereby ensuring stable combustion of the 2 A arc.

Fig. 12 shows the radial distribution of T_e and T_h at the four positions in Fig. 11. As shown in Fig. 11(a), at the center of the plane 0.1 mm away from the cathode tip, the T_e and T_h are 11,370 K and 9274 K, respectively. T_h decreases by 95.1 % in the region 3 mm away from the arc axis, whereas the T_e decreases by only 60.9 %, resulting in an increase in the ratio of T_e to T_h from 1.22 to 9.8. Thus, the deviation from LTE becomes stronger when moving away from the arc axis. A similar phenomenon occurs elsewhere in the arc column. Beyond the radius of 3.5 mm, the T_e slowly decreases only with thermal conductivity due to the loss of Joule heating, resulting in the decrease of θ . In general, the θ value is larger at the arc fringe. It has been shown that the conductivity decreases rapidly with the increase of θ at the same T_e [31]. Thus, the θ higher than 4 outside the radius of 3 mm make the conductivity of these regions extremely low, resulting in almost no current passing through these places. Therefore, with the deviation from the arc axis, the θ gradually increases, along with the decrease of T_e , resulting in a rapid decline in conductivity. This is what causes nearly all the current to be concentrated near the arc axis, resulting in the formation of a columnar arc.

4.3. Effect of current on arc characteristics

For various arc currents, the distributions of T_e and T_h along the arc axis are shown in Fig. 13. A significant increase in T_e with arc current was observed only in two regions within 2 mm from the cathode and anode. In other regions, the T_e remains nearly constant at 7600 K for the 1–4 A arcs. However, the T_h decreases from 4550 K to 3280 K as the current decreases from 4 A to 1 A, revealing that the non-equilibrium phenomena are more pronounced for lower current arcs. It is well

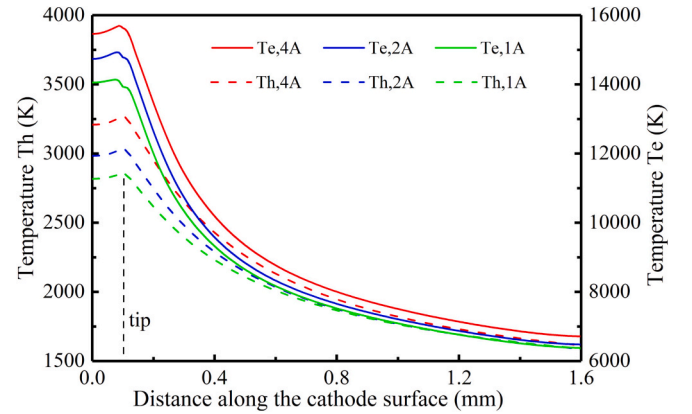


Fig. 15. Distributions of T_e and T_h along the cathode surface for various arc currents.

known that the T_e plays a much greater role than the T_h in maintaining plasma conductivity [30]. Although T_h of the 1 A arc column is much lower than that of the 4 A arc, its T_e can remain roughly the same as that of the 4 A arc. This is why 1 A arc can maintain high conductivity and a stable burning arc.

In order to maintain the high T_e in the arc column, the low-current arc must shrink the arc column to obtain a sufficiently high current density, as shown in Fig. 14. The radius of the low-current arc column $R_{a(x)}$ is expressed as [40]:

$$R_{a(x)} = \left(\frac{I}{\pi E_x \sigma_{x,r=0}} \right)^{0.5} \quad (28)$$

where E_x and $\sigma_{x,r=0}$ are the electric field strength and conductivity on the arc axis, respectively. On the arc axis ($z = 12$ mm), the current density increases from 3.59×10^5 A/m² to 4.09×10^5 A/m² as the current increases from 1 A to 2 A, but only increases to 3.79×10^5 A/m² as the current increases to 4 A, as shown in Fig. 14. Thus, the current density on the arc axis is not proportional to the arc current. This is the result of the self-contraction of the low-current arc to maintain its combustion under the premise of ensuring energy conservation. When the current of the

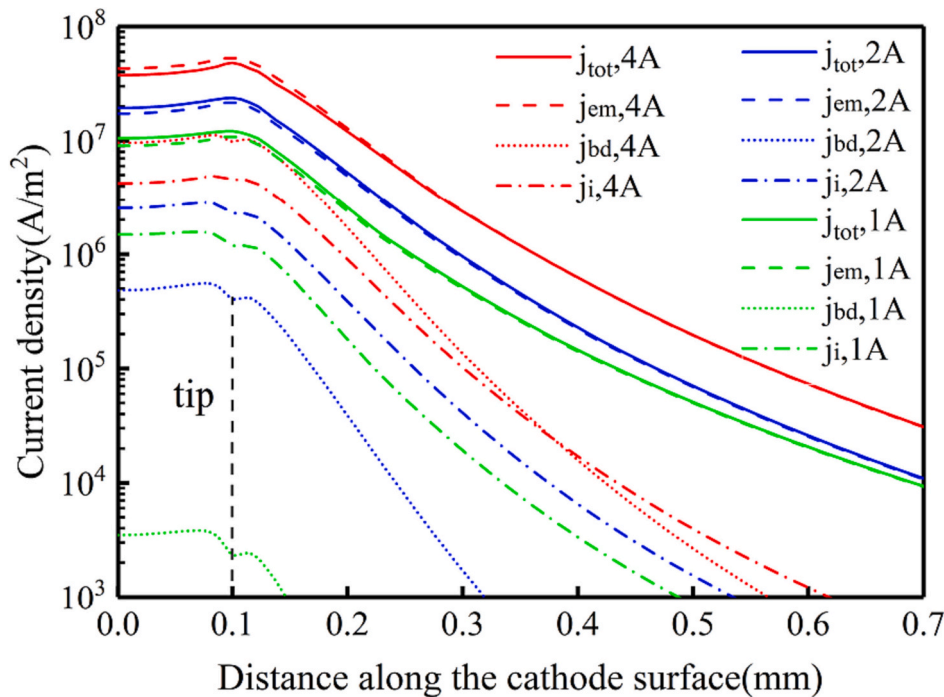


Fig. 16. Distributions of all components of the arc current density along the cathode surface for various arc currents.

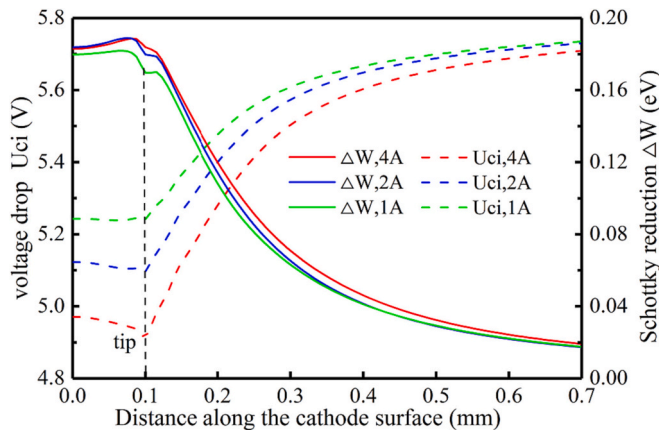


Fig. 17. Distributions of U_{ci} and ΔW along the cathode tip for various arc currents.

micro-arcs drops, the energy input to the arc column is also reduced. However, the arc column needs to maintain a high T_e . In this case, only reducing the side area of the arc column can effectively reduce the energy loss to maintain energy conservation. As a result, when the current is reduced from 4 A to 1 A, the side area of the arc column was minimized, causing the arc to contract into a cylindrical shape with a radius of only 0.9 mm. Such contraction also reduces arc stability, as discussed in our previous work [3].

Distributions of parameters along the cathode surface are illustrated in Figs. 15–17. As can be seen in Fig. 15, the temperature distribution along the cathode surface is non-monotonic. The hottest part of the cathode is the edge of the front surface, where heat removal via thermal conduction is the worst. This is quite different from the distribution of T_e along the front surface of the cathode, which has a maximum at 0.019 mm from the edge. This is because the edge of the front surface has the best conditions for thermal-conduction heat removal for T_e . As the current decreases from 4 A to 1 A, the temperature of the cathode tip edge decreases from 3270 K to 2860 K, while the T_e decreases from

15,600 K to 13,900 K. In addition, the T_h and T_e of all current arcs drop rapidly along the lateral surface. It is noteworthy that the radius of the cathode tip plateau is sufficiently small (0.1 mm), allowing even a 1 A arc to generate a temperature of 2860 K.

The distribution of electric current density distribution closely resembles that of T_h distribution, with the exception that the former, being heavily dependent on T_h and T_e , experiences a steeper decline along the lateral surface (Fig. 16). For 1–4 A arcs, j_{em} accounts for >86.2 % of the total current j_t , while the j_{bd} and j_i have a very low proportion due to high U_c . That is, the current at the cathode surface is mainly determined by thermionic emission. The U_c increases with the decrease of the arc current, resulting in a rapid decrease in the electron density and ion current diffused to the cathode. Due to the sharp cathode shape, the maximum current density of the cathode tip in micro-TIG welding is much greater than that in HID lamps for the same current [20]. Thus, only a lower U_c is needed in micro-TIG welding to generate enough heat flux to heat the cathode tip for maintaining electron emission. This factor is also responsible for the considerably lower U_c observed in

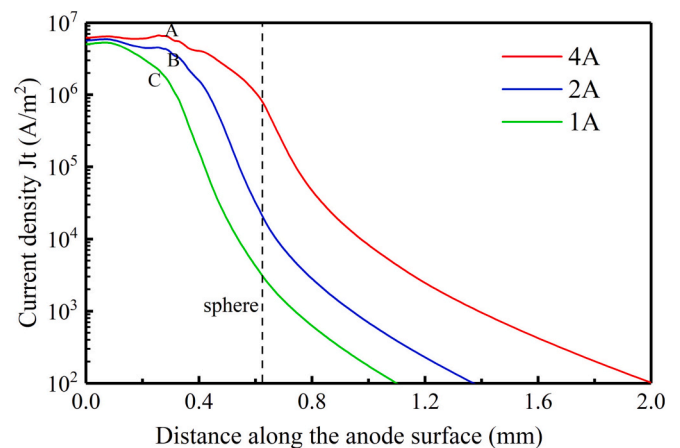


Fig. 18. Distributions of current density j_{ea} along the anode surface for various arc currents.

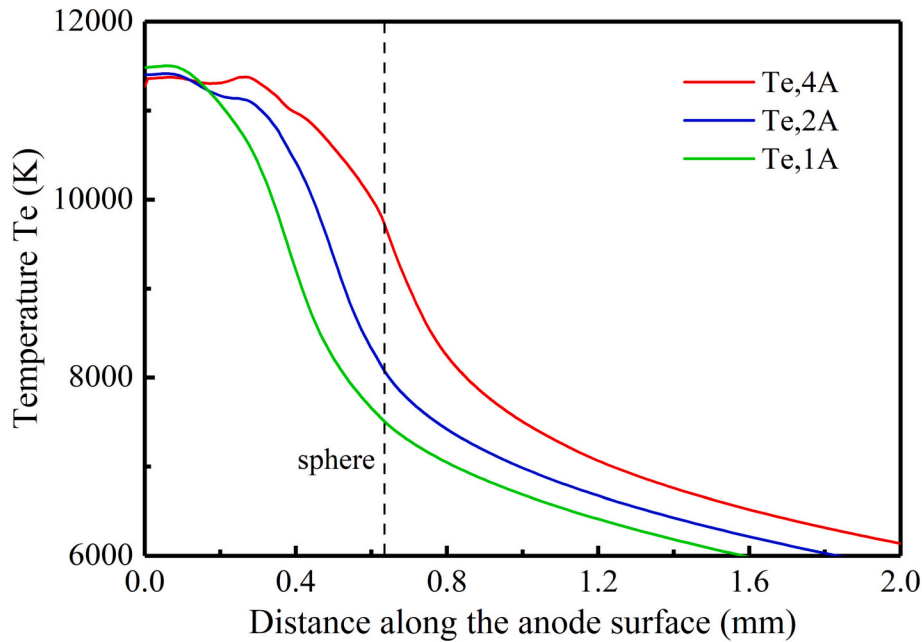


Fig. 19. Distributions of T_e along the anode surface for various arc currents.

micro-TIG welding, as compared to HID lamp. In addition, sufficiently high heat flux is also responsible for the rapid stabilization of the arc after ignition in micro-TIG welding [3].

$$n_{es} = n_{ei} e \left(\frac{eU_{ci}}{k_B T_e} \right) \quad (29)$$

In Fig. 17, the distributions of U_{ci} and ΔW on the cathode surface explains why the current density is more concentrated at the cathode tip plateau. The U_{ci} at the cathode tip plateau is lower than that on the lateral surface, allowing more ions to bombard the cathode tip plateau (Eq. (29)). This causes the temperature of the cathode tip plateau to rise more easily. Moreover, the ΔW at the cathode tip plateau is greater than that at the lateral surface, making the cathode tip plateau more favorable for electron emission. Therefore, in micro-TIG welding, it is crucial to finely polish the cathode tip into a sharp point to enhance its thermal electron emission capabilities, thereby improving arc stability.

The important physical parameters on the anode surface are analyzed in Figs. 18–19. The j_{ea} of 1–4 A arcs are close to 5.8×10^6 A/m² within the distance of 0.1 mm from the center of the hemispherical surface of the anode. Outside of this region, the j_{ea} of 1 A arc decreases rapidly, while that of 2 and 4 A arcs decreases rapidly only outside the region 0.25 mm from the center, and the j_{ea} of 2 A arc drops faster than that of 4 A arc. The distribution of the T_e along the anode surface follows a pattern similar to that of the j_{ea} , and this depends on the dimensions of the arc column. As the current increases, the arc column expands, resulting in a significant contraction near the anode hemispherical surface. Under the influence of radial current density, the j_{ea} peaks at positions “A” and “B” for the 4 A and 2 A arcs, respectively. Moreover, the j_{ea} of the 4 A arc is much larger than that of the 1 A and 2 A arcs beyond 0.25 mm from the center. It is easy to deduce that this difference is very important in the welding process because the heat flux input to the anode ($q_a = j_{ea}(2k_B T_e + W_a)/e - q_{rad}$) is mainly determined by j_{ea} and T_e . In other words, as the current increases, the heating area on the anode surface expands rapidly. In practical micro-TIG welding, a thinner copper wire is wrapped around a workpiece in micro-TIG welding. To prevent the copper wire from burning, it is often necessary to utilize an appropriate low-current arc. Therefore, understanding the distributions of j_{ea} and T_e on the anode surface through simulation has paramount significance for improving the welding quality of the micro workpiece.

5. Conclusions

- (1) A semi-empirical 2-T arc model is established to investigate the low-current arc characteristics in micro-TIG welding. By coupling the plasma to the near cathode layer, the 2-T arc model can well simulate the volt-ampere characteristics of a low-current arc and the contraction of the arc column with decreasing current.
- (2) The low-current arc column is cylindrical, with slight expansions in the near cathode and anode regions. The thermal non-equilibrium characteristics of plasma are very significant along the axis, but there is an obvious drop in the near cathode and anode regions due to the high-temperature gas. In the radial direction, the thermal non-equilibrium characteristic gradually increases as it deviates from the arc axis, leading to a sharp decline in conductivity and the size of the cylindrical arc column.
- (3) When the current in a low-current arc decreases, the arc deviates further from thermal equilibrium, causing a significant decrease in the temperature of heavy particles, and leading to the contraction of the arc column. However, the electrons on the arc axis still have a high temperature to maintain high conductivity and a stable arc.
- (4) In micro-TIG welding, the current on the cathode surface mainly consists of the electrons emitted during the thermionic process. In comparison to HID lamps, the model reveals that the sharper cathode in micro-TIG welding leads to higher heat flux density and lower near-cathode voltage drop. These findings are consistent with the experimental measurement.
- (5) The 2-T arc model can predict the current density distribution on the surface of the micro workpiece, revealing that the main heating zone expands as the arc current increases. In other words, the heating and melting behavior of the anode surface varies with the arc current.

Declaration of competing interest

This work was funded by the National Natural Science Foundation of China (51175184). No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. All data generated or analysed during this study are included in this published article. The code for the model is written according to Fluent's

user manual. The manuscript does not address ethical issues.

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